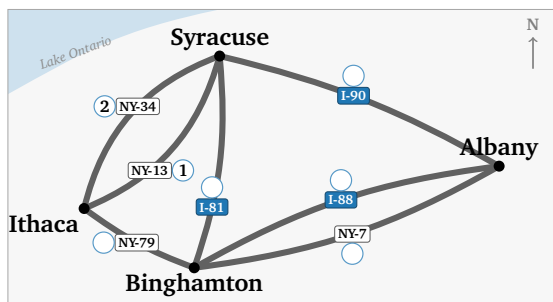


## Part 1 — Seven highways

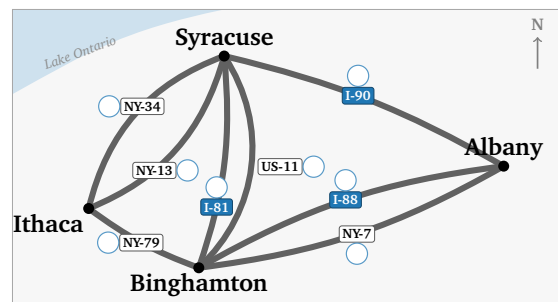
Seven highways join four Upstate New York cities: Ithaca, Syracuse, Binghamton, and Albany.

**Question 1(a):** Drive every highway exactly once, without lifting your pencil. Start and finish anywhere; cities may repeat, highways may not. Write in each circle when you drive that road — 1 first, then 2, on to 7. Two are filled in: you start at Ithaca, take NY-13, then NY-34 straight back.

**Question 1(b):** The same four cities, and an eighth highway: **US-11**, which really does run alongside I-81. Number your route again. Can you drive every highway without lifting your pencil?



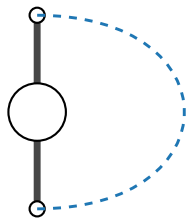
1(a): seven highways



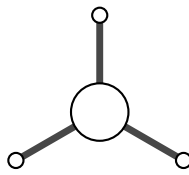
1(b): with US-11

**Question 2:** One city, on its own. Arriving uses up one highway, leaving uses up another — so highways get used in **pairs**.

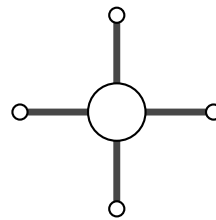
Join two road-ends with a curved line to mean “in here, out there”. Pair up as far as you can. The first is done.



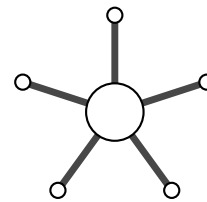
2 roads



3 roads



4 roads



5 roads

|                 | 2<br>roads | 3<br>roads | 4<br>roads | 5<br>roads |
|-----------------|------------|------------|------------|------------|
| pairs you drew  | 1          |            |            |            |
| roads left over | 0          |            |            |            |

- (a) Which of these cities can you drive straight *through*?
- (b) What do those numbers share?
- (c) A left-over road has no partner. What must this city be for the tour?

**Question 3:** Count the highways at each city. Ithaca is done.

|                | I | S | B | A | how many<br>have a<br>left-over? |
|----------------|---|---|---|---|----------------------------------|
| seven roads    | 3 |   |   |   |                                  |
| with US-11 too |   |   |   |   |                                  |

One start, one finish, so at most how many cities may have a left-over road?

For the seven roads: can you drive every highway exactly once? If so, where does that drive begin and end?

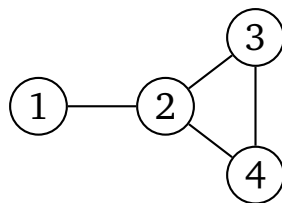
For the eight roads, with US-11: can you drive every highway exactly once?

Why is there no route with eight?

## Part 2 — Three kinds of route

---

A different network: four stops, numbered 1 to 4.



**Question 4(a):** Fill in the table. The last column takes one of these three names. First row done.

Repeats nothing: **path**.    Never repeats a line: **trail**.    Repeats both: **walk**.

| route     | repeats a line? | repeats a stop? | name |
|-----------|-----------------|-----------------|------|
| 1 2 3 2 4 | yes (2–3 twice) | yes (2)         |      |
| 1 2 3 4 2 |                 |                 |      |
| 1 2 3 4   |                 |                 |      |

**Question 4(b):** Write a route from 1 to 2 that repeats a stop but no line.

**Question 4(c):** Back to the highway map. Your drive in 1(a) used no road twice, but it did come back to a city it had already visited. Which is it — a path, a trail, or a walk?

**Question 4(d):** It could not have been a path. How many different cities would a path of 7 roads have to touch? How many does the highway map have? Why does that rule out a path?

## Part 3 — Predictions

---

Part 4 hands you a notebook — a computer page that will run the network from Part 2 for you. Work the answers out on paper first; the notebook will check them.

**Question 5(a):** Write 1 in row  $i$ , column  $j$  if a line joins  $i$  and  $j$ , else 0, for the network from Part 2. Row 1 done.

|       |   |   |   |   |
|-------|---|---|---|---|
| $A =$ | 1 | 2 | 3 | 4 |
| 1     | 0 | 1 | 0 | 0 |
| 2     |   |   |   |   |
| 3     |   |   |   |   |
| 4     |   |   |   |   |

**Question 5(b):** Add up each row, then compare with the network from Part 2. What are the four row totals? What does each row total count, for that node?

**Question 5(c):** A **2-step route** goes along one line, then along one more. From node 1 to node 3,  $1 \rightarrow 2 \rightarrow 3$  is one such route. Using the network from Part 2, count the 2-step routes:

How many go from 1 to 3? Write one out.

How many go from 1 to 2? Why so few?

How many go from node 2 back to node 2?

**Question 5(d):** The notebook in Part 4 will square your grid for you. Before it does, say what you expect to find in  $A^2$ .

What do you expect in row 1, column 3? In row 1, column 2?

What is on the diagonal of  $A^2$ , and why?

## Part 4 — The lab, on your own

---



**Do this one by yourself**, with this sheet beside you. The notebook takes the same four cities and seven highways you have been drawing on, and asks you to write them down in a form a computer can read.

**Point a camera at the square.** If that fails, type [go.skojaku.com/m01lab](http://go.skojaku.com/m01lab)